

Production Generative AI Master Curriculum

Comprehensive 21-Module Engineering Syllabus & Stack (2026)

CORE LEARNING MODULES (1 - 20)

Module 1: Introduction to GenAI

- **Concepts:** Foundation Models, LLMs vs Traditional AI, Open vs Closed Source, Tokens & Tokenization.
- **Parameters:** Context Window, Temperature, Top-k, Top-p, Streaming, Pricing & Selection.
- **Models:** GPT, Claude, Gemini, Llama, Qwen, Mistral, Gemma, Phi, DeepSeek.

Module 2: Prompt Engineering

- **Anatomy:** System, User, and Assistant Prompts.
- **Techniques:** Role, XML, Markdown, Structured (JSON output), and Function prompting.
- **Reasoning:** Zero-shot, Few-shot, Chain of Thought (CoT), Self-consistency, Prompt Optimization & Templates.

Project: AI Travel Planner

Module 3: LLM APIs

- **Providers:** OpenAI, Anthropic, Google Gemini, Groq, Together AI, OpenRouter.
- **Features:** Chat Completion, Streaming, Tool/Function Calling, JSON Mode, Structured Output, Image/Audio/Vision input.

Project: Universal LLM Chat Application

Module 4: Embeddings

- **Theory:** Vector embeddings, Similarity Search, Cosine Similarity, Dot Product, Dense vs Sparse retrieval, Hybrid search.
- **Libraries:** sentence-transformers, OpenAI embeddings.

Project: Semantic Search

Module 5: Vector Databases

- **Systems:** FAISS, Chroma, Pinecone, Qdrant, Weaviate, Milvus.
- **Operations:** Indexing, Metadata filtering, Collections, Upsert, Delete, Search, ANN search, Distance metrics.

Project: Document Search Engine

Module 6: Advanced RAG

- **Pipeline:** Chunking strategies, Chunk overlap, Metadata, Query rewriting, Context compression, Re-ranking.
- **Strategies:** Parent-child, Multi-vector, Hybrid retrieval, Citation generation.
- **Loaders:** PDF, DOCX, CSV, HTML, Markdown, SQL, APIs.

Projects: Chat with PDFs, Corporate KB, Documentation Bot

Module 7: LangChain Ecosystem

- **Core:** Chains, Prompt Templates, Messages, Memory management, Output Parsers.
- **Execution:** Retrievers, Tools, Agents, Callbacks, LCEL (LangChain Expression Language).

Project: Multi-tool AI Assistant

Module 8: LlamaIndex Ecosystem

- **Architecture:** Document loaders, Nodes, Indexes, Query engines, Chat engines.
- **Routing:** Response synthesis, Router query engine.

Project: Enterprise RAG

Module 9: AI Agents

- **Cognition:** Planning, Reasoning, Reflection, ReAct frameworks.
- **Mechanics:** Tool selection, Multi-step execution, Memory systems, Context management.
- **Frameworks:** LangGraph, CrewAI, OpenAI Agents SDK, AutoGen.

Projects: Coding/Research Agents, Customer Support

Module 10: Model Context Protocol (MCP)

- **Spec:** MCP Architecture, Client vs Server, Resources, Prompts, Tools, Capabilities.
- **Infr:** Transport (stdio, HTTP), Authentication.
- **Integrations:** GitHub, Gmail, PostgreSQL, Filesystem, Slack, Frappe ERPNext.

Project: Personal MCP Server

Module 11: Tool Calling

- **Mechanics:** Function calling, JSON schema validation, Tool execution, Multi-tool workflows.
- **Execution:** Parallel & Sequential tools, Error handling, Input verification.

Projects: Weather, SQL, and Calendar Assistants

Module 12: Memory Systems

- **Types:** Short-term, Long-term, Conversation, Vector, Episodic, and Semantic memory.
- **Frameworks:** LangGraph Memory, Mem0, Zep.

Project: Personal AI Assistant

Module 13: Multimodal AI

- **Domains:** Vision models, OCR, Image/Audio/Video understanding, Speech-to-text, Text-to-speech.
- **Models:** Whisper, GPT-4o Vision, Gemini Vision, Qwen-VL.

Projects: Image Q&A, Invoice Reader

Module 14: Fine-Tuning

- **Methods:** SFT (Supervised Fine-Tuning), PEFT, LoRA, QLoRA, DPO (Direct Preference Optimization), Preference tuning.
- **Data:** Synthetic data generation.
- **Libraries:** TRL, PEFT, Unsloth.

Module 15: AI Evaluation

- **Metrics:** Hallucination detection, Faithfulness, Relevance, Correctness, Toxicity, Safety, Latency, Cost.
- **Frameworks:** DeepEval, Ragas, LangSmith, OpenAI Evals.

Module 16: LLM Observability

- **Tools:** LangSmith, Phoenix, Helicone, LangFuse.
- **Topics:** Tracing, Logs, Metrics, Prompt versioning, Cost monitoring.

Module 17: AI Security & Guardrails

- **Threats:** Prompt Injection, Jailbreaks, Data leakage, PII exposure.
- **Defenses:** Secrets management, Output filtering, API Guardrails.
- **Libraries:** Guardrails AI, NeMo Guardrails.

Module 18: Backend & Deployment

- **Backend:** FastAPI, Async Python, Streaming APIs, Authentication.
- **Infra:** Docker, Kubernetes (basics), Nginx, Redis.
- **Cloud:** AWS, Azure, Google Cloud, Render, Railway.

Module 19: Local LLMs

- **Tools:** Ollama, LM Studio, vLLM, llama.cpp.
- **Optimization:** Quantization, GGUF formats, GPU vs CPU inference, Local model serving.

Module 20: Production AI Architecture

- **Gateway:** AI Gateway, Semantic Caching, Load balancing, Rate limiting.
- **Reliability:** Cost optimization, Retry strategies, Asynchronous streaming, Background jobs, Message Queues (Celery/RabbitMQ), Enterprise logging/monitoring.

MODULE 21: CAPSTONE PROJECTS

- ChatGPT Clone
 - PDF Chatbot (RAG)
 - Multi-document RAG System
 - SQL AI Assistant
 - GitHub Code Reviewer
 - AI Research Agent
 - AI Email Assistant
 - Voice Assistant
- Vision-based Document Analyzer
 - Multi-Agent Workflow
 - MCP Server for PostgreSQL
 - MCP Server for ERPNext/Frappe
 - AI Customer Support Agent
 - Enterprise Knowledge Assistant
 - AI Coding Copilot

RECOMMENDED LEARNING STACK (2026)

Category	Technologies
Programming	Python
LLM APIs	OpenAI, Anthropic, Gemini, Groq, OpenRouter
Prompting	Prompt Engineering, Structured Outputs
Embeddings	Sentence Transformers, OpenAI Embeddings
Vector DB	Qdrant, Pinecone, Chroma, FAISS
RAG	LangChain, LlamaIndex
Agents	LangGraph, CrewAI, OpenAI Agents SDK
MCP	MCP Python SDK, MCP Inspector, Custom MCP Servers
Memory	Mem0, Zep, LangGraph Memory
Evaluation	Ragas, DeepEval, LangSmith
Observability	LangFuse, Phoenix, Helicone
Deployment	FastAPI, Docker, Redis, Nginx
Local LLMs	Ollama, vLLM, llama.cpp
Security	Guardrails AI, NeMo Guardrails

STRUCTURED LEARNING SEQUENCE

1. Generative AI & LLM Fundamentals
2. Prompt Engineering

3. LLM APIs & Structured Outputs
4. Embeddings
5. Vector Databases
6. RAG (basic → advanced)
7. LangChain & LlamaIndex Ecosystems
8. Tool Calling & Function Integration
9. AI Agents & LangGraph Frameworks
10. Model Context Protocol (MCP) Implementation
11. Memory Systems & Multimodal AI
12. Fine-Tuning Frameworks
13. Evaluation, Observability & Guardrails
14. Local LLM Quantization & Serving
15. Deployment & Production AI Architecture
16. End-to-End Capstone Projects